Three Janko geometries and their residues

Human Compression project: bounded application

We apply the accepted shadow and residue criterion to the literal J_1 inputs supplied with Jacobs and Leemans, *An algorithmic analysis of the intersection property*, LMS J. Comput. Math. 7 (2004), 284–299. The complete common proof is retained in *Shadows and residues in finite coset geometries*; this application supplies its actual hypotheses and finite decisions. It does not repeat the earlier complete subgroup census.

1 Fixed inputs and group actions

The supplementary files `j1.rk3` and `j1.rk4` contain one and two active entries, respectively. There is no final permutation of those construction entries. Write A for the rank-three entry and B, C for the rank-four entries in that literal order. Their generators are complete permutations of the same 266 points. Each family of maximal parabolic generators generates the same simple group of order 175560. An explicitly computed isomorphism transports it to the previously accepted J_1 model. The following subgroup orders are computed, not used to infer geometry.

Entry	Borel order	ordered maximal-parabolic orders
A	4	$(120, 120, 660)$
B	2	$(120, 60, 660, 120)$
C	2	$(60, 660, 660, 120)$

For every fixed-type set J, the residue group is H_J, with remaining parabolics $H_J \cap H_i$. Every eligible flag extension satisfies the exact orbit equality in the common theorem. Every rank-one coset action has degree at least two, and its point stabilizer has two orbits. In each positive-rank residue some type action is primitive; in each residue of rank at least two the parabolic join is the whole group. Both directional shadow tests hold in every rank-two residue. These computations verify, separately, full flag transitivity, firmness, local two-transitivity, residual weak primitivity, residual connectedness and $(\mathrm{IP})_2$.

A second input identification needs no new census. Under the ambient isomorphism, matching the entire parabolic families under automorphisms and type permutations identifies A, B, C with entries $27, 31, 32$ of the previously exhausted phase-six J_1 catalogue. They are exactly its entries of rank at least three satisfying local two-transitivity. The matching procedure tries every type permutation, then successively maps each subgroup while restricting to the stabilizer of all subgroups already placed. It therefore proves equivalence of the entire family, not merely agreement of subgroup orders. Prior catalogue completeness remains an explicit retained input to this identification; it is not re-established by the present shadow calculation.

2 All residual shadow properties

Theorem 1. *The truth values for the three fixed archive entries are as follows. A dash indicates that the corresponding rank exceeds the input rank.*

Entry	$(\mathrm{IP})_2/(\mathrm{WIP})_2$	$(\mathrm{IP})_3/(\mathrm{WIP})_3$	$(\mathrm{IP})_4/(\mathrm{WIP})_4$	*full IP/WIP*
A	*yes/yes*	*no/no*	—	*no/no*
B	*yes/yes*	*yes/yes*	*yes/yes*	*yes/yes*
C	*yes/yes*	*no/no*	*no/no*	*no/no*

Proof. The preceding checks establish every hypothesis of the common residue isomorphism and shadow recognition theorem. For every proper type subset J, form $K = H_J$ and the parabolics $K \cap H_i$. For each remaining target type, enumerate all normalized object–flag double cosets and all possible witness types and stabilizer cosets. A positive answer covers every required orbit. A negative answer records its actual normalizing permutation and failed intersection after exhausting the entire witness search for that pair. The rank-two shortcut is the independently proved neighborhood-intersection criterion, including complete bipartite residues and excluding comparison of an object with itself.

The exact run covers 37 base residues including panels. Its 26 residues of rank at least two produce 65 directional type decisions. At rank k, conjunction over every type in every base residue gives $(\mathrm{IP})_k$. For $(\mathrm{WIP})_k$ a successful type may be chosen separately in each residue. The complete saved decisions give the displayed table. The full properties are the conjunction over $2 \leq k \leq$ rank; rank one is automatic and rank zero is excluded. □

The actual rank-two residue counts represented by our base flags are $3192, 77539, 70224$ for A, B, C. The rank-three counts for B, C are $6118, 4921$. Each is a sum of the actual indices $[G : H_J]$, without assuming that different fixed-type sets have isomorphic residues.

3 Source correspondence and retained limits

The aggregate rank-three, rank-four and full columns agree with the J_1 rows of Tables 1, 3 and 7. Tables 2, 4 and 8 call the positive rank-four geometry 4.1 in the earlier classification. The supplied permutations and the prior phase-six identifiers do not themselves identify that historical name. We therefore record B as the positive literal archive entry without silently assigning a published label. The unavailable classification chapter is not newly acquired or recovered.

The common orbit algorithm and the published shadow results belong to Jacobs and Leemans. The generic covering and rank-two proofs, the complete phase-six census and its independent validation are retained accepted work. Our task is their bounded residue-level application to three literal archived inputs. GAP 4.15.1 group, coset and automorphism routines remain trusted computational premises; this is not a complete Lean proof. Other sporadic groups and exact historical labels lie outside this packet. There is no new original, question or compression credit and no measurement.